

NITESH KUMAR

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Skills

Languages: Python, JavaScript, TypeScript, Java

Frameworks/Libraries: Node.js, Express.js, Socket.IO, Pandas, NumPy, Scikit-learn, LangChain, Streamlit

Tools/Platforms: Git, VS Code, PostgreSQL, Prisma, MongoDB, Redis, Excel, Power BI

Soft Skills: Problem-Solving, Teamwork, Time Management

Projects

SwiftChat – Real-Time Chat Application | [GitHub](#) | [Live Demo](#)

Aug 2026 – Present

- Architected a **real-time chat system** using **3 microservices** (API, Socket, Consumer) to enhance scalability and modularity.
- Reduced **message delivery latency** by **70%** using **Socket.IO + Redis Pub/Sub** for instant real-time communication and presence tracking.
- Built an **event-driven pipeline** with **Apache Kafka** handling **1000+ messages/min** throughput, ensuring reliable and zero-loss message processing.
- Improved **system performance** by **60%** by decoupling database writes via Kafka and asynchronous consumer processing with **Prisma ORM**.

Tech Stack: Express.js, TypeScript, React.js, Socket.IO, Kafka, Redis Pub/Sub, PostgreSQL

EquityBot – AI Research Assistant | [GitHub](#) | [Live Demo](#)

Aug 1 – Aug 20, 2026

- Built a **research-automation tool** that answers company- and sector-specific queries with cited sources, cutting the manual effort of secondary industry research.
- Designed an end-to-end **retrieval pipeline** (LangChain + FAISS) for document ingestion, indexing, and semantic search over financial news, converting unstructured web content into a searchable knowledge base.
- Developed the **ingestion layer** to pull and parse articles directly from financial-news URLs/APIs, keeping the research base current with minimal manual effort.

Tech Stack: Python, LangChain, FAISS

Intrusion Detection System using Machine Learning | [GitHub](#)

May 2025

- Developed an **Intrusion Detection System** to classify network traffic anomalies using the 41-feature **NSL-KDD benchmark dataset**.
- Conducted **EDA** (correlation heatmaps, class distributions, statistical t-tests) and applied **Label Encoding** to preprocess categorical features.
- Trained a **Random Forest** classifier with balanced class weights, achieving **99.76% validation accuracy** on a held-out split of the training data.
- Deployed an interactive **Streamlit** web app supporting batch CSV prediction, single-record inference, and performance-metric visualization.

Tech Stack: Python, NumPy, Pandas, Scikit-Learn, Streamlit

Training

Training in Machine Learning – CipherSchools | [Certificate](#)

Jun 2025 – Jul 2025

- Completed a **6-week** structured training covering data science workflow, Python (Pandas, NumPy), data visualization (Matplotlib, Seaborn, Plotly), and Power BI.
- Applied data preprocessing, statistical analysis, and machine learning techniques (**Regression, Decision Trees, Random Forest, SVM, K-Means, PCA**).
- Built projects including a chatbot (NLP) and an image classifier (CNN).

Certifications

NPTEL – Social Networks — [Certificate](#)

Jul 2025 – Nov 2025

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate — [Certificate](#)

Aug 2025 – Sep 2025

Gen AI NASSCOM Certification – SFJ Skill Development Program — [Certificate](#)

Feb 2025

Achievements

- Solved **100+ problems** on [LeetCode](#), [HackerRank](#), and [GeeksforGeeks](#).
- Earned a **4-star rating** in Java on HackerRank.

Education

Lovely Professional University — B.Tech CSE; CGPA: 8.19

Aug 2023 – Present

Doon Valley Public School, Nalagarh, Himachal Pradesh — Intermediate; Percentage: 83.6%

Apr 2022 – Mar 2023

Doon Valley Public School, Nalagarh, Himachal Pradesh — Matriculation; Percentage: 78.6%

Apr 2020 – Mar 2021